CHAPTER 14 · REFERENCE DEPTH

Call Signs & Establishing Communication

On a busy frequency a dozen aircraft may be listening at once, so every transmission must say who is calling and who is being called — unambiguously, every time. Call signs are that addressing system, and the rules around them are exact: which suffix names which controller, how an airline flight becomes a call sign, when you may shorten one, and what you must always read back.

SYLLABUS MAP

Part III (i) Call signs · aeronautical & aircraft call signs · establishment, continuation & transfer · read-back · test procedures

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◎ Learning objectives — by the end of this chapter you will be able to...

- Identify aeronautical (ground) station call signs by their suffix.
- Form the three types of aircraft call sign and apply the abbreviation rules correctly.
- Make a correct initial call and continue communication.
- Handle a transfer of communication.

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- State the items that must always be read back.
- Carry out a radio-check (test) procedure.

14.1 Why call signs

14.2 Ground-station call signs

14.3 The three aircraft call-sign types

14.4 Abbreviation rules

14.5 Establishing communication

14.6 Transfer of communication

14.7 Mandatory read-back items

14.8 Test (radio-check) procedure



The ATC tower is the hub of the airport's communication. Ground station suffixes like TOWER, GROUND, and DELIVERY explicitly define who you are talking to.

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14.1 Why call signs

FIRST PRINCIPLES — ADDRESSING ON A PARTY LINE

VHF is a shared, simplex "party line": everyone hears everything. A call sign is the **address** that tells one specific station "this is for you" and tells everyone else "this is not for me". Every transmission therefore names **who is being called** and **who is calling**, so there can never be doubt about who must act.

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14.2 Ground-station call signs — the suffix names the service

IN PLAIN TERMS

A ground station's call sign is the **place name + a suffix**, and the suffix tells you which service you have reached.

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Ground Station Suffixes

Suffix	Service	Suffix	Service
DELIVERY	Clearance delivery	CONTROL	Area control centre

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Suffix	Service	Suffix	Service
GROUND	Surface movement control	**RADAR**	Radar (general)
TOWER	Aerodrome control	**DIRECTOR**	Approach radar (final director)
APPROACH	Approach control	**PRECISION**	Precision approach radar (PAR)
DEPARTURE	Departure control	**INFORMATION**	Flight Information Service (FIS)
APRON	Apron management	**RADIO**	Aeronautical station (HF/communications)

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MNEMONIC — THE GROUND FLOW

A departing flight typically walks through **Delivery** → **Ground** → **Tower** → **Departure/Radar** → **Control**; an arrival comes back **Control** → **Approach/Director** → **Tower** → **Ground**. Learn the suffixes in that running order.

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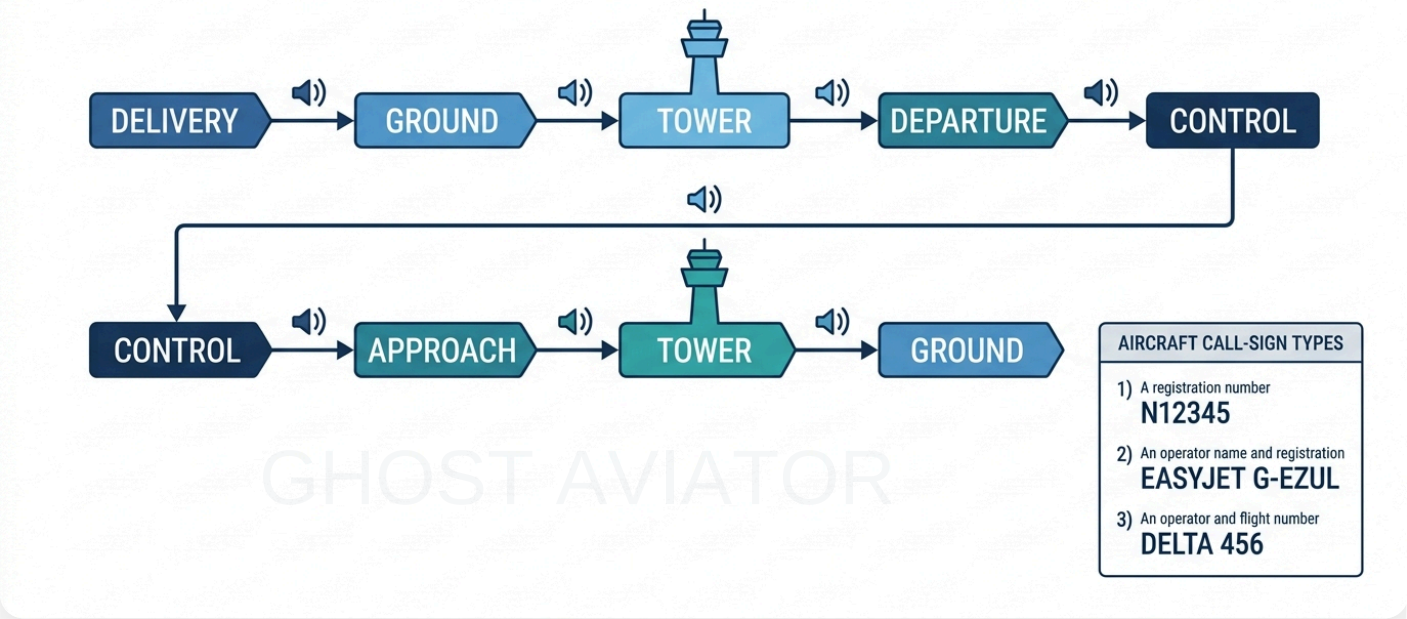


Figure 14.1 — Who you talk to — the ground-station flow.

14.3 The three aircraft call-sign types

✈ TYPE 1 — REGISTRATION / TYPE

The aircraft's **registration** spelled phonetically: **VT-ABC** → "Victor Tango Alfa Bravo Charlie". A general-aviation aircraft may prefix the manufacturer/type for the first call: "Cessna Victor Tango Alfa Bravo Charlie".

✈ TYPE 2 — OPERATOR + LAST CHARACTERS OF REGISTRATION

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The telephony designator of the operator followed by the last characters of the registration: e.g. operator "GHOSTAIR" + registration ...ABC → "Ghostair Alfa Bravo Charlie".

✈ TYPE 3 — OPERATOR + FLIGHT NUMBER

The operator's telephony designator followed by the **flight identification** (number/letters): "Air India one zero one" for AIC101. This is the airline call sign you hear most. The flight number is spoken **digit by digit**.

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The three types of aircraft call signs ensure there is a clear, standard way to address both small general aviation aircraft and large commercial airliners.

⚠ EXAM TRAP

The telephony designator is **not** the IATA two-letter code; "AI" is written but spoken as the telephony name "**Air India**". And the flight number is read digit by digit — "one zero one", not "one hundred and one".

14.4 Abbreviation rules

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✓ WHEN AND HOW A CALL SIGN MAY BE SHORTENED

1. An aircraft call sign may be **abbreviated only after the ground station has first abbreviated it** — never on the pilot's initiative, and not at all if there is any risk of confusion with another similar call sign on frequency.
2. **Type 1** (registration) abbreviates to the **first character and at least the last two**: "VT-ABC" → "Victor Tango … Bravo Charlie" or "V…BC".
3. **Type 2/3** (operator + flight number) is **not** abbreviated — the flight number is kept in full.

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🎙 TRANSCRIPT — FULL THEN ABBREVIATED

A/C Delhi Ground, Victor Tango Alfa Bravo Charlie, request taxi.

ATC Victor Tango Alfa Bravo Charlie, taxi to holding point Alfa One. (ground initiates full)

A/C Taxi holding point Alfa One, Victor Tango Bravo Charlie. (now abbreviated)

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14.5 Establishing communication

THE INITIAL CALL

An initial call names **(1) who you are calling**, then **(2) who you are**, then keeps the first message brief. On a first contact you may state a short request; for a longer message, wait to be invited ("pass your message").

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TRANSCRIPT — ESTABLISHING COMMUNICATION

A/C Delhi Approach, Ghostair Alfa Bravo Charlie.

ATC Ghostair Alfa Bravo Charlie, Delhi Approach, pass your message.

A/C Ghostair Alfa Bravo Charlie, Cessna 172, 15 miles north, FL080, inbound for landing, information Charlie.

ATC Ghostair Alfa Bravo Charlie, roger, descend FL060, QNH one zero one three.

A/C Descend FL060, QNH one zero one three, Ghostair Alfa Bravo Charlie.

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COCKPIT REALITY — LISTEN BEFORE YOU CALL

Before that first call, listen out for a few seconds: you may learn the runway in use, the QNH and how busy the controller is, and you avoid transmitting over someone mid-clearance.

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14.6 Transfer of communication

BEING HANDED ON

As you pass between sectors, ATC **transfers** you to the next frequency: "contact [station] on [frequency]". You read back the frequency, change over, and make a fresh initial call on the new frequency. "**Monitor**" means tune and listen out without calling immediately.

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TRANSCRIPT — TRANSFER

ATC Ghostair Alfa Bravo Charlie, contact Delhi Tower one one eight decimal one.

A/C Delhi Tower one one eight decimal one, Ghostair Alfa Bravo Charlie.

A/C Delhi Tower, Ghostair Alfa Bravo Charlie, FL060 inbound. (new call on new frequency)

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14.7 Mandatory read-back items

WHY READ-BACK EXISTS

A read-back closes the loop: the controller hears you repeat the instruction and can catch a mishearing before it becomes an incident. Certain safety-critical items **must always** be read back in full.

ALWAYS READ BACK

- **ATC route clearances**
- clearances to **enter, land on, take off from, backtrack, cross or hold short of** any runway
- **runway in use**
- **level instructions**
- **heading** and **speed** instructions
- **altimeter (QNH/QFE)** settings
- **SSR (squawk) codes**
- **frequency changes**
- VDF information and transition levels.

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MNEMONIC

"Runways, Levels, Headings, Speeds, QNH, Squawk, Frequencies" — if it's one of these, read it back in full. End the read-back with your call sign.

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A pilot's mental checklist for read-backs ensures no safety-critical instruction is ever glossed over.

14.8 Test (radio-check) procedure

TRANSCRIPT — RADIO CHECK

A/C Delhi Tower, VT-ABC, radio check on one one eight decimal one.

ATC VT-ABC, Delhi Tower, readability five.

A/C VT-ABC, readability five also, thank you.

REPORTING READABILITY

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State the readability on the **1–5 scale** (1 unreadable … 5 perfectly readable). If readability is poor, the receiving station may say "readability three, you are broken/distorted", prompting a frequency or equipment check.

☆ Numbers to memorise

Essential Facts for Chapter 14	
Fact	**Value**
Ground suffixes	Delivery · Ground · Tower · Approach · Departure · Director · Radar · Control · Information
Call-sign types	1: registration · 2: operator + reg · 3: operator + flight number
Abbreviation	Only after ATC abbreviates; Type 1 → first + last two; never abbreviate the flight number
Initial call	Who you call → who you are → brief message / request
Transfer	"Contact [station] [freq]" → read back freq → new call
Always read back	Runways, levels, headings, speeds, QNH, squawk, frequency changes
Radio check	Readability 1–5

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Question bank

Part A — MCQs (click an option to check)

1. The ground-station suffix "GROUND" denotes:

Area control

Surface movement control

Approach control

Flight information

Answer: Surface movement control.

GROUND = surface movement control (taxi); TOWER = aerodrome control.

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2. "DIRECTOR" is associated with:

Apron management

Approach radar (final vectoring)

Clearance delivery

Area control

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Answer: Approach radar (final vectoring).

Director provides final approach radar vectoring.

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3. The suffix for Flight Information Service is:

CONTROL

RADAR

INFORMATION

TOWER

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Answer: INFORMATION.

INFORMATION denotes the Flight Information Service.

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4. An airline flight AIC101 is spoken as:

"A-I-C one hundred one"

"Air India one zero one"

"AI one zero one"

"Air India one hundred and one"

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Answer: "Air India one zero one".

Telephony designator + flight number digit by digit: "Air India one zero one".

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5. An aircraft call sign may be abbreviated:

Whenever the pilot wishes

Only after the ground station has first abbreviated it

Never

Only on initial contact

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Answer: Only after the ground station has first abbreviated it.

The pilot may abbreviate only once ATC has done so, and not if confusion is possible.

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6. Type 1 call sign VT-ABC abbreviates to:

"VT-A"

First character and at least the last two — e.g. "Victor … Bravo Charlie"

"ABC"

It cannot be abbreviated

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Answer: First character and at least the last two.

Registration call signs shorten to the first plus at least the last two characters.

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7. An airline (operator + flight number) call sign is abbreviated by:

Dropping the operator name

Shortening the flight number

It is not abbreviated — the flight number is kept in full

Using the IATA code

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Answer: It is not abbreviated.

Operator + flight-number call signs are not abbreviated.

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8. An initial call should give, in order:

Your request, then your call sign

Who you are calling, then who you are

The frequency, then your message

Your altitude, then your heading

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Answer: Who you are calling, then who you are.

Name the station being called first, then your own call sign.

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9. "Pass your message" means:

Change frequency

Go ahead with your message / request

Standby

Read back

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Answer: Go ahead with your message / request.

It invites the aircraft to transmit its full message.

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10. On a transfer of communication, ATC says:

"Squawk ident"

"Contact [station] on [frequency]"

"Report ready"

"Cleared to land"

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Answer: "Contact [station] on [frequency]".

A transfer instructs you to contact the next unit on a stated frequency.

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11. "Monitor [frequency]" means:

Call immediately

Tune and listen out, without calling immediately

Squawk the code

Change runway

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Answer: Tune and listen out, without calling immediately.

MONITOR = listen out on the frequency without an immediate call.

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12. Which of the following must always be read back?

Traffic information

A runway clearance (enter/land/take off/cross/hold short)

A weather report

A position report

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Answer: A runway clearance.

Runway clearances are mandatory read-back items.

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13. A squawk code is:

Not read back

A mandatory read-back item

Only read back at night

Read back as a whole number

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Answer: A mandatory read-back item.

SSR codes must be read back, digit by digit.

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14. A read-back should normally end with:

"Over and out"

Your call sign

"Roger"

The frequency

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Answer: Your call sign.

Finish the read-back with your own call sign for positive identification.

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15. A radio check reply of "readability three" means the signal is:

Perfectly readable

Readable but with difficulty

Unreadable

Readable

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Answer: Readable but with difficulty.

3 = readable but with difficulty (1 unreadable, 5 perfectly readable).

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16. The telephony designator spoken for an airline is:

The IATA two-letter code

The assigned telephony name (e.g. "Air India")

The aircraft registration

The flight number only

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Answer: The assigned telephony name.

The spoken designator is the telephony name, not the written two-letter code.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **What does the suffix of a ground station's call sign tell you? Give examples.**

Model Answer:

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It identifies the service: Delivery (clearance), Ground (surface movement), Tower (aerodrome control), Approach, Departure, Director (final radar), Control (area), Information (FIS).

▼ ? **Describe the three types of aircraft call sign.**

Model Answer:

Type 1 — the registration spelled phonetically (optionally with the type); Type 2 — the

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operator's telephony designator plus the last characters of the registration; Type 3 — the operator's telephony designator plus the flight number, spoken digit by digit.

▼ ? When may an aircraft abbreviate its call sign, and how?

Model Answer:

Only after the ground station has first abbreviated it, and not if confusion with another call sign is possible. A registration call sign shortens to the first character and at least the last two; operator-plus-flight-number call signs are not abbreviated.

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▼ ? What information does an initial call contain?

Model Answer:

Who you are calling, then who you are, then a brief message or request; for a longer message you wait for "pass your message".

▼ ? List the items that must always be read back.

Model Answer:

ATC route clearances; runway clearances (enter/land/take off/backtrack/cross/hold short); the runway in use; level, heading and speed instructions; altimeter settings; SSR codes; and frequency changes.

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▼ ? How do you carry out a radio check?

Model Answer:

Call the station with "radio check on [frequency]"; the station replies with your readability on the 1–5 scale, and you acknowledge with your own readability assessment.

⚡ 60-SECOND REVISION CARD

Ground suffix = service:

Delivery·Ground·Tower·Approach·Departure·Director·Control·Information.

Call signs: 1 registration · **2** operator+reg · **3** operator+flight number (digit by digit).

Abbreviate only after ATC does; Type 1 → first + last two; never shorten the flight number.

Initial call: who you call → who you are → brief message. Transfer: "contact [stn] [freq]".

Always read back: runways, levels, headings, speeds, QNH, squawk, frequencies; end with your call sign. Radio check = readability **1–5**.

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